

PEDRO H. R. DE ANDRADE

Computer Engineering
de Andrade, P. H. R.

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ACADEMIC SUMMARY

I'm currently a Computer Engineering student with a passion for astronomy and data science. I have experience in scientific research, with a focus on computational astrophysics, and a history of publications in national conferences. I'm always seeking to learn and contribute to the scientific community, exploring new technologies and methodologies to advance our understanding of the universe. I have a hobby in animations and simulations, which allows me to create engaging visualizations to communicate complex concepts in an accessible way, with a focus on science outreach of physics-related content.

ACADEMIC BACKGROUND

Bachelor in Computer Engineering
Fluminense Federal Institute *Campus Bom Jesus do Itabapoana*
August. 2023 – present Rio de Janeiro, Brazil

Technical Degree in Informatics
Fluminense Federal Institute *Campus Bom Jesus do Itabapoana*
February. 2020 – December. 2022 Rio de Janeiro, Brazil

RESEARCH PROJECTS

Analysis of Dynamical Simulations of Merging Galaxy Clusters
Fluminense Federal Institute *Campus Bom Jesus do Itabapoana*
September. 2022 – March. 2023 Rio de Janeiro, Brazil
Junior Scientific Initiation Scholarship - CNPq
Supervisor: Dr. Ana Cecília Soja

Simulating the Impact of Satellites on Astronomical Observations
Fluminense Federal Institute *Campus Bom Jesus do Itabapoana*
2025 – present Rio de Janeiro, Brazil
Scholarship in the Scientific Initiation project - CNPq
Supervisor: Dr. Ana Cecília Soja

MobFog at IFFMaker
Fluminense Federal Institute *Campus Bom Jesus do Itabapoana*
August. 2023 – February. 2024 Rio de Janeiro, Brazil
Scholarship in the Scientific Initiation project - IFF
Supervisor: Dr. Ana Cecília Soja

Anomaly Detection in Stars of the Milky Way: Exploring Data from Gaia and Other Surveys with Machine Learning
Fluminense Federal Institute *Campus Bom Jesus do Itabapoana*
2024 – present Rio de Janeiro, Brazil
Scholarship in the Scientific Initiation project - CNPq
Supervisor: Dr. Ana Cecília Soja and Dr. Maria Luiza Linhares Dantas

EXTERNAL COURSES

Networks
Fluminense Federal Institute *Campus Bom Jesus do Itabapoana*
August. 2020 Rio de Janeiro, Brazil

Operational Systems
Fluminense Federal Institute *Campus Bom Jesus do Itabapoana*
August. 2021 Rio de Janeiro, Brazil

Server Administration
Fluminense Federal Institute *Campus Bom Jesus do Itabapoana*
August. 2022 Rio de Janeiro, Brazil

RESEARCH INTERESTS

EVENTS

Science Fairs

Analysis of dynamical simulations of merging galaxy clusters

Poster – 21st Brazilian Science and Engineering Fair (FEBRACE)

2023 • São Paulo, Brazil  Understanding dark matter from extragalactic shocks

Poster – 40th International Science and Technology Fair (MOSTRATEC)

2023 • Novo Hamburgo, Brazil  Simulating the impact of satellites on astronomical observations

Poster – IX Brazilian Science Initiation Fair (FEBIC)

2024 • Pomerode, Brazil  Anomaly detection in stars of the Milky Way

Poster – XVI Scientific and Technological Exhibition of the Açai Institute

2025 • Belém, Brazil

Conferences

Analysis of dynamical simulations of merging galaxy clusters

Oral Presentation – XV Scientific and Technological Initiation Congress and IX Fluminense Congress of Postgraduate Studies (CONFICT)

2024 • Campos dos Goytacazes, Brazil  Analysis of dynamical simulations of merging galaxy clusters

Oral Presentation – XVI Scientific and Technological Initiation Congress and X Fluminense Congress of Postgraduate Studies (CONFICT)

2025 • Campos dos Goytacazes, Brazil  GALACTIC Catalog: stellar properties and chemistry

Poster – XLVIII Annual Meeting of the Brazilian Astronomical Society (SAB)

2025 • Caxambu, Brazil  GALACTIC Catalog: stellar properties and chemistry

Oral Presentation – XVII Scientific and Technological Initiation Congress and XI Fluminense Congress of Postgraduate Studies (CONFICT)

2026 • Campos dos Goytacazes, Brazil  Anomaly detection in stars of the Milky Way

Poster – 78th Annual Meeting of the Brazilian Society for the Progress of Science (SBPC)

2026 • Niterói, Brazil  Anomaly detection in stars of the Milky Way

Poster – Winter School of Astrophysics

2026 • Rio de Janeiro, Brazil  Anomaly detection in stars of the Milky Way

Poster – XLIX Annual Meeting of the Brazilian Astronomical Society (SAB)

2026 • Natal, Brazil  MobFog at IFFMaker

Presentation – Congress of Teaching, Research and Extension (CONEPE)

2026 • Campos dos Goytacazes, Brazil

Academic Weeks

Analysis of dynamical simulations of merging galaxy clusters

Participant – VIII Knowledge Showcase and I Job Fair

2020 • Bom Jesus do Itabapoana, Brazil  Understanding dark matter from extragalactic shocks

Participant – IX Knowledge Showcase and II Job Fair

2021 • Bom Jesus do Itabapoana, Brazil  Simulating the impact of satellites on astronomical observations

Participant – I Week of Computer Engineering and Management

2021 • Bom Jesus do Itabapoana, Brazil  Understanding dark matter from extragalactic shocks

Speaker – X Knowledge Showcase and III Job Fair

2022 • Bom Jesus do Itabapoana, Brazil  Simulating the impact of satellites on astronomical observations

Participant – II Week of Computer Engineering and Management

2022 • Bom Jesus do Itabapoana, Brazil  MobFog at IFFMaker

Poster – XI Knowledge Showcase and IV Job Fair

2023 • Bom Jesus do Itabapoana, Brazil  MobFog at IFFMaker

Participant – III Week of Computer Engineering and Management

2023 • Bom Jesus do Itabapoana, Brazil  MobFog at IFFMaker

Participant – XII Knowledge Showcase and V Job Fair

2024 • Bom Jesus do Itabapoana, Brazil  MobFog at IFFMaker

Speaker and Organizer – IV Week of Computer Engineering and Management

2024 • Bom Jesus do Itabapoana, Brazil  MobFog at IFFMaker

Participant – XIV Knowledge Showcase and VII Job Fair

2026 • Bom Jesus do Itabapoana, Brazil

Proceedings

[1] P. H. R. de Andrade, M. L. L. Dantas, and A. C. Soja. « Stellar properties and chemical features of the Gaia Catalogue of Nearby Stars observed by GALAH DR4 ». In: *Boletim da Sociedade Astronômica Brasileira*. Vol. 37. 1. 2026, pp. 308–309.

[2] Pedro Henrique Rocha de Andrade, Ana Cecília Soja, and Maria Luiza Linhares Dantas. « How varied are stellar populations in the Solar vicinity? Exploring GCNS and GALAH DR4 with t-SNE ». in: *Boletim da Sociedade Astronômica Brasileira*. Natal, RN, 2026.

[3] Pedro Henrique Rocha de Andrade, Ana Cecília Soja, and Maria Luiza Linhares Dantas. « Propriedades estelares e características químicas do Gaia Catalogue of Nearby Stars observado pelo GALAH DR4 ». In: *Anais do XVI Congresso de Iniciação Científica e Tecnológica e X Congresso Fluminense de Pós-Graduação*. Campos dos Goytacazes, RJ: UENF, 2026.

[4] Pedro Henrique Rocha de Andrade, Ana Cecília Soja, and Maria Luiza Linhares Dantas. « Detecção de Anomalias em Estrelas da Via Láctea: Explorando Dados do Gaia e Outros Surveys com Aprendizado de Máquina ». In: *Anais da 78ª Reunião Anual da Sociedade Brasileira para o Progresso da Ciência (SBPC)*. Niterói, RJ: UFF, 2026.

[5] Pedro Henrique Rocha de Andrade, Ana Cecília Soja, and Maria Luiza Linhares Dantas. « Detecção de Anomalias em Estrelas da Via Láctea: Explorando Dados do Gaia e Outros Surveys com Aprendizado de Máquina ». In: *Resumos da Escola de Inverno de Astrofísica do Observatório Nacional*. Rio de Janeiro, RJ: Observatório Nacional, 2026.

[6] Maycon Jorge Deláqua da Silva, Pedro Henrique Rocha de Andrade, and Ana Cecília Soja. « Simulando o impacto de satélites em observações astronômicas ». In: *Anais do XVI Congresso de Iniciação Científica e Tecnológica e X Congresso Fluminense de Pós-Graduação*. Campos dos Goytacazes, RJ: UENF, 2025.

[7] Maycon Jorge Deláqua da Silva et al. « Simulando o impacto de satélites em observações astronômicas ». In: *XVI Mostra de Ciência e Tecnologia do Instituto Ação (MCTIA)*. 2025.

[8] Maycon Jorge Deláqua da Silva et al. « Simulando o impacto de satélites em observações astronômicas ». In: *Anais do XVI Congresso de Iniciação Científica e Tecnológica e IX Congresso Fluminense de Pós-Graduação*. Campos dos Goytacazes, RJ: UENF, 2024.

[9] Maycon Jorge Deláqua da Silva et al. « Simulando o impacto de satélites em observações astronômicas ». In: *Anais da IX FEBIC – Feira Brasileira de Iniciação Científica*. 2024. ISBN: 978-65-991144-6-5.

[10] Pedro Henrique Rocha de Andrade and Ana Cecília Soja. « Análise de simulações dinâmicas de aglomerados de galáxias em fusão ». In: *Anais da 21ª FEBRACE – Feira Brasileira de Ciências e Engenharia*. Vol. 21. 2023, p. 5. ISBN: 978-65-89190-18-9.

[11] Pedro Henrique Rocha de Andrade and Ana Cecília Soja. « Entendendo a matéria escura a partir de choques extragaláticos ». In: *Anais da XI Mostra do Conhecimento 2023*. 2023, p. 17.

[12] Pedro Henrique Rocha de Andrade, Ana Cecília Soja, and Fabrício Barros Gonçalves. « Entendendo a matéria escura a partir de choques extragaláticos ». In: *Anais da MOSTRATEC 2023*. 2023, p. 102.

[13] Gabriel Coelho de Souza Poubel et al. « MobFog no IFFMaker ». In: *Anais da XI Mostra do Conhecimento 2023*. 2023, p. 21.

Articles

[14] P. H. R. de Andrade, M. L. L. Dantas, and A. C. Soja. « Stellar properties and chemical features of the Gaia Catalogue of Nearby Stars observed by GALAH DR4 ». In: *arXiv e-prints* (Dec. 2025). DOI: [10.48550/arXiv.2512.10037](#).

AWARDS


IX FEBIC

3rd Place in the Undergraduate Category with the work "Simulating the Impact of Satellites on Astronomical Observations"


XVI MCTIA

1st Place in the Undergraduate Category with the work "Simulating the Impact of Satellites on Astronomical Observations"

STATISTICS


3 Oral Presentations

National Events


7 Posters

National Events


9 Science Fairs/Congresses

National Events


10 Academic Weeks

Regional Events

PROGRAMMING

 Python

 Git

 Github

 LaTeX

 C++

 Java

 NodeJS

LANGUAGES

Portuguese



English



French



Spanish

